Animation-Based Framework for Speech Production and Perception Experiments

Akos Buza

July 2026

1 Introduction

This repository provides a framework for browser-based speech production and perception experiments using animated visual stimuli. It was originally developed for research on information structure (from now on IS), but is designed to be language-independent and easily adaptable to other research questions.

The framework combines animations of simple two-dimensional shapes, participant interfaces, audio recording, and experimental control within a single *jsPsych*-based environment (de Leeuw et al., 2023). It supports a complete experimental workflow, from stimulus presentation and response collection to the generation of speech recordings for acoustic and perceptual analyses.

The paradigm is easy to adopt, and requires only minimal modification to adapt to a new language, or for a variety of IS categories. In most cases, only the participant instructions and linguistic materials need to be translated, while the experimental logic, animations, recording interface, and data structure remain unchanged. This facilitates reproducible cross-linguistic research and enables comparable experiments across different languages.

The framework supports read-speech production for maximum control, spontaneous production for "real-world" speech data, and also hybrid designs. Speech recordings collected during production experiments can be used directly as stimuli for perception experiments, providing a unified material and workflow for both production and perception research.

Experiments can be conducted locally or deployed online through platforms such as *JATOS* (Lange, Kühn & Filevich, 2015) and *Prolific*. All responses, data, and recordings are stored in a format suitable for analysis in *R*, *Python*, or other software.

The framework builds upon the animation-based elicitation methodology developed by Beth Ann Hockey (1998), extending it into a reusable platform for integrated production and perception experiments in cross-linguistic research. This previous work demonstrated that simple animated visual events provide an effective means of eliciting spontaneous and cross-linguistically comparable speech. Hockey's implementation was developed specifically to investigate the realization and interpretation of focus in English and Hungarian. The present framework generalizes this methodology into a reusable, browser-based experimental platform, for all categories of IS. It is modular, language-independent, supports online data collection, integrates production and perception workflows, and is designed to accommodate a wide range of experimental paradigms. The demo experiments included in the repository aim to demonstrate that such a paradigm can easily account for IS categories including (but not limited to): broad focus, narrow focus, new information focus, contrastive focus, corrective focus, givenness, mirativity. Potentially other questions in experimental linguistics and speech science can also be investigated using the animated tasks.

2 Features

- Browser-based experiments implemented in *jsPsych*
- Language-independent, animation-based tasks
- Production and perception experiments within a common framework
- Read-speech, spontaneous speech, and hybrid elicitation modes
- Automatic audio recording with playback and re-recording option
- Modular trial structure requiring minimal programming to adapt
- Customizable instructions, prompts, and experimental materials
- Automatic collection of responses, reaction times, metadata, and recordings
- Compatible with local deployment, *JATOS*, *MindProbe*, and *Prolific*
- Structured output for acoustic, phonetic, and statistical analyses

3 Repository Structure

The repository is organized into a small number of independent components that can be adapted separately. In most cases, researchers only need to translate the linguistic materials and instructions while the experimental logic and visual stimuli remain unchanged.

`production-index-*.html` Production experiments for English, German, and Hungarian, including participant instructions, animations, recording interfaces, and the complete experimental workflow.

`perception-*.html` Perception experiments for English, German, and Hungarian. These present recorded speech together with visual stimuli and collect participant judgments.

`audio-extract-animation-task.py` Python utility for extracting recorded audio files from the experimental JSON output.

`README.md` Repository overview, installation instructions, screenshots, and basic usage information.

`ANIMATION_TASK_DOCUMENTATION_GITHUB_VERSION.pdf` Comprehensive documentation describing the framework, experimental paradigms, customization, deployment, and analysis workflow.

`CITATION.cff` Citation metadata for referencing the repository.

`LICENSE` MIT license governing the use and redistribution of the framework.

The production and perception experiments share the same underlying architecture, making it straightforward to adapt the framework to new languages or experimental paradigms by modifying the participant instructions and linguistic materials while preserving the experimental logic.

4 Elicitation Types

The production framework supports three elicitation types that differ only in the type of linguistic material presented to participants. The animations, recordings, and experimental workflow remain otherwise unchanged.

4.1 Read Speech

Participants view an animated event, read the context question, then read a predefined target sentence during recording. This provides maximal control and is well suited for acoustic analyses and stimulus creation.

4.2 Spontaneous Speech

Participants view the same animation but receive only the context question. They are free to choose their own elicitation.

4.3 Hybrid Designs

Intermediate designs are also supported. Researchers may combine target sentences, elicitation questions, contextual information, or general task instructions as required by the experimental design.

5 Animation Design

The framework intentionally employs simple geometric objects characterized by only three visually salient properties: Shape, colour, and size. This minimizes cognitive load while providing multiple independent dimensions along which discourse can be manipulated. Using the same visual scene, researchers can introduce the shape as new information (What entered the scene?), the colour as new information (What colour is the square?), the size as new information (Which object is larger?), contrast alternative objects or properties (Which shape is green?), elicit corrective responses (The square is green, isn't it?), or investigate mirativity (What happened? — The circle ate the square.). This way, a relatively small set of reusable animations can support a wide range of IS configurations while preserving the same underlying visual event.

6 Customizing the Paradigm

The framework was designed to minimize the amount of programming required to create new experiments. In most cases, adapting the paradigm only requires modifying the participant instructions. Additional experimental conditions can be added as needed.

Depending on the experimental design, researchers can customize:

- participant instructions;
- target sentences and elicitation prompts;
- animations and visual stimuli;
- trial structure and experimental flow;
- recording parameters;
- participant interface.

7 Data Output

The framework saves data in a format that is suitable for analysis in *R*, *Python* etc.

The production experiment generates trial-level metadata. It stores the corresponding audio recordings, participant identifiers, experimental conditions, timing information, and responses. Perception experiments store participant responses, reaction times, stimulus identifiers, and experimental conditions.

The framework can also record additional behavioral measures, such as replay counts, response revisions, trial completion times, and other custom interaction events defined by the researcher.

This enables easy processing, including, but not limited to: forced alignment, manual annotation, acoustic analysis, and statistical modeling.

This output format allows production and perception datasets to be combined within a single research workflow and integrated with existing analysis tools.

For further information please read up on jsPsych (de Leeuw, J.R., Gilbert, R.A., & Luchterhandt, B. 2023).

8 Long-Term Vision

The long-term goal of this project is to provide a standardized, open-source framework for combined speech production and perception experiments. By combining reusable animations, a common experimental architecture, and standardized data output, the framework aims to simplify experiment development while promoting methodological consistency across studies.

Although the example experiments focus on IS, the framework is intended to support a much wider range of research questions in linguistics, psycholinguistics, speech science, and experimental phonetics. Its language-independent design also facilitates cross-linguistic research. In principle, adapting the experiment requires only changes to the linguistic materials rather than the underlying experimental infrastructure.

Ultimately, the project aims to provide a reusable research platform that encourages replication, collaboration, and the development of new experimental paradigms.

9 Citation

If you use this framework in your research, please cite the associated publication once it becomes available.

Until then, please cite the GitHub repository:

> Buza, Á. (in preparation). *Animation-Based Framework for Speech Production and Perception Experiments*. GitHub repository.

Users are encouraged to cite any accompanying methodological or empirical publications relevant to their use of the framework. These will be listed in future releases.

10 License

This project is released under the MIT License.

Users are free to use, modify, distribute, and adapt the framework for academic, educational, or commercial purposes, provided that the original copyright notice and license are retained.

The animations, source code, and accompanying documentation may therefore be incorporated into new experimental paradigms, extended for additional research questions, or adapted to other languages without requiring separate permission from the author.

A copy of the complete license text is included in the `LICENSE` file distributed with the repository.

11 Audio Extraction Utility

The repository includes a Python utility that extracts audio recordings from the JSON output generated by the experiments. Recordings are stored as Base64-encoded strings and are automatically decoded and saved as individual audio files using the corresponding `recording_name`. The extracted files can then be used for forced alignment, acoustic analysis, or as stimuli in perception experiments.

12 Production Experiment Conditions

Table 1 summarizes the example conditions included in the production experiment. Animations present a visual event, while the accompanying context question specifies the intended context and prompts the desired IS category (e.g., broad focus, new information focus, contrastive focus, or givenness). Participants evaluate or produce utterances as responses to a particular question under discussion rather than to the animations alone. Naturally, if your research question requires getting rid of questions under discussions, you are free to do that.

The example production experiments illustrate how different IS categories can be elicited by systematically manipulating the discourse context while keeping the underlying animations largely unchanged. By varying context questions, the same visual events can instantiate categories such as broad focus, new information focus, narrow focus, contrastive focus, corrective focus, mirative focus, as well as different configurations of givenness, newness, and background. Researchers can readily adapt these contextual manipulations to investigate other information-structural phenomena. The presented conditions are just an example inventory. Conditions can be added or deleted, as needed.

13 Perception Experiments

The repository also includes a set of examples for speech perception experiments. Recordings collected during production experiments can be used directly as perception stimuli, enabling integrated production–perception studies: The results of the production experiment directly inform and serve the perception part as well.

Each perception trial combines a visual stimulus (e.g., an animation, image, or written context), one or more replayable audio recordings, and an associated response task. As in the production experiments, they are fully modular and can be easily customized the same way as described above.

13.1 Supported Response Paradigms

The current implementation includes several commonly used response paradigms:

Likert ratings Participants rate recordings on a discrete scale (e.g., 1–5).

Continuous slider ratings Participants provide judgments on a continuous scale.

Forced-choice tasks Participants compare two or more recordings and select the preferred or more appropriate response.

Ranking tasks Participants rank multiple recordings using drag-and-drop, sequential selection, or custom interfaces.

Different response paradigms can be freely combined within the same experiment. The framework includes several commonly used measures for convenience, although *jsPsych* (de Leeuw et al., 2023) supports a much wider range of experimental paradigms.

13.2 Response Validation

The framework also supports custom response validation. Researchers can enforce mandatory responses, prevent invalid rankings (e.g., duplicate ranks), or implement any experiment-specific validation rules before allowing participants to proceed.

13.3 Hosting and Data Storage with JATOS

The framework is compatible with **JATOS** (*Just Another Tool for Online Studies*; Lange, Kühn, & Filevich, 2015) and has been tested on the **MindProbe** server (https://mindprobe.eu/), although it can be deployed on any institutional or locally hosted JATOS installation.

13.4 Running Locally

1. Clone or download the repository.
2. Start a local web server.
3. Open the html in your browser.
4. Grant microphone access when prompted.

13.5 Running with JATOS

Upload the experiment files as a JATOS study component, set `index.html` as the entry point, and launch the study using a test or participant link.

Acknowledgments

I would like to thank Mark Liberman for introducing me to the work of Beth Ann Hockey. His recommendation motivated the development of the present work by adopting it to my interests in research methods.

References

[1] Hockey, B. A. (1998). *The Interpretation and Realization of Focus: An Experimental Investigation of Focus in English and Hungarian*. Ph.D. Dissertation, University of Pennsylvania. ProQuest Dissertations and Theses, No. 9829914.

[2] de Leeuw, J. R., Gilbert, R. A., & Luchterhandt, B. (2023). *jsPsych: Enabling an Open-Source Collaborative Ecosystem of Behavioral Experiments. Journal of Open Source Software*, *8*(85), 5351. `https://doi.org/10.21105/joss.05351`

[3] Lange, K., Kühn, S., & Filevich, E. (2015). *"Just Another Tool for Online Studies" (JATOS): An Easy Solution for Setup and Management of Web Servers Supporting Online Studies. PLOS ONE*, *10*(6), e0130834. `https://doi.org/10.1371/journal.pone.0130834`

[4] Palan, S., & Schitter, C. (2018). *Prolific.ac—A Subject Pool for Online Experiments. Journal of Behavioral and Experimental Finance*, *17*, 22–27. `https://doi.org/10.1016/j.jbef.2017.12.004`

Table 1: Example information-structural conditions included in the production experiment. These are illustrative examples of how the framework can be used. Researchers are free to adapt the discourse contexts and specify the information-structural categories according to their preferred theoretical framework.

Information Structure	**Prompt**	**Target Sentence**
Broad Focus	What happened?	A triangle entered the scene.
New information focus	What entered the scene?	A square entered the scene.
New information focus	What entered the scene?	A circle entered the scene.
Narrow Focus	Which shape pushed the other one out of the scene?	The square pushed the triangle out of the scene.
Broad Focus	What happened?	The square jumped.
Contrastive Focus	Which shape jumped, the circle or the square?	The square jumped.
Broad Focus	What happened?	The circle grew larger, and the square became smaller.
New information focus on colour	What colour is the square?	The square is red.
Contrastive Focus	Which shape is green, the circle or the square?	The circle is green.
Corrective Focus	The square is green, isn't it?	No. The circle is green.
Mirative Focus	What happened?	The circle ate the square.
Corrective Focus	The square ate the circle, didn't it?	No. The circle ate the square.